

Chapter 6 Review

Learning Objectives

- Consolidate knowledge of all emerging technologies covered in Chapter 6
- Analyze the interconnected nature of modern technology ecosystems
- Evaluate emerging technologies critically with technical and ethical awareness

Engage (5-7 minutes)

Play “Technology Connection”: students draw a random emerging technology card (AI, IoT, cloud, blockchain, 5G) and must explain how it connects to at least three other technologies. For example: IoT generates data, 5G transmits it, cloud processes it, AI analyzes it, blockchain secures it. This reveals that these technologies form an interconnected ecosystem, not isolated innovations.

Explore (10-12 minutes)

Students complete a comprehensive concept map linking all Chapter 6 topics: AI → machine learning → applications → IoT → sensors → cloud → infrastructure → blockchain → security → 5G → connectivity → ethics → responsibility. They identify critical connections and potential conflicts. Then they solve three scenario-based problems that require knowledge from multiple chapters.

Explain (10-12 minutes)

Chapter 6 introduced the technologies reshaping our world. AI and machine learning enable intelligent systems; IoT connects physical devices; cloud computing provides scalable infrastructure; blockchain offers decentralized trust; 5G delivers the connectivity these technologies demand. Together, they form a technology stack where each layer enables the others. Critical evaluation requires understanding both capabilities and limitations, benefits and risks, potential and ethical boundaries. These technologies are not theoretical—they are deployed now and students will work with them throughout their careers.

Elaborate (8-10 minutes)

Cross-cutting analysis: examine a smart city project and identify which Chapter 6 technologies are involved at each level. Discuss technology adoption lifecycles—how innovations move from early adopters to mainstream. Address the digital divide: who benefits from these technologies and who is left behind? Future-gazing: what emerging technologies not yet covered might define the next decade?

Evaluate (5-8 minutes)

Comprehensive final assessment: a multi-part examination covering definitions and concepts (recall), analysis of technology scenarios (application), ethical evaluation of a technology deployment (critical thinking), and a short essay on the technology most likely to transform the student's future career (synthesis and reflection). Students also complete a course self-assessment identifying their growth areas and setting learning goals for future studies.